CATERPILLAR®

DEALER BEST PRACTICE SERIES

Oil Drain Interval Extension

Application	Maintenance	Site Management	Component Rebuild	**Component Life Management**	MARC Management

December 2006
1206-2.1-1050

1.0 Introduction

Hydraulic and powertrain oil quality is a critical factor in component durability. Changing oil at the recommended interval is intended to remove contaminants and maintain oil viscosity and additive levels to specification. However, some progressive customers and dealers have found that keeping oil very clean reduces both contaminant levels and reduces the rate of oil oxidation. Viscosity and additive levels also last much longer when oil is kept clean. This behavior led to experimentation to see if oil life could be extended, which has proven successful at numerous sites throughout the world.

2.0 Best Practice Description

In general, oil life may be extended if two major factors are controlled and closely monitored:

- Contaminant levels
- Oxidation

Contaminant levels can be controlled by aggressive filtration and maintaining oil cleanliness levels of ISO18/15 or better. This may be achieved by the use of off-board filtration (kidney-looping), improved onboard filtration, or both.

Oxidation of the oil is caused by overheating, chemical reaction with wear metals in the oil, or a combination of both. Overheating (cooking) the oil is caused by excessive operating temperatures. Chemical oxidation is caused by a chemical reaction between wear metals in the oil and the oil itself. Minimizing wear metals dramatically slows the chemical oxidation process.

3.0 Implementation Steps

There are no formal recommendations from Caterpillar on extending oil life for powertrain and hydraulic components. Some informal recommendations have emerged from dealer/ customer experience include:

1. On new machines or rebuilt components, a larger than normal amount of break-in debris is generated. A good practice is to change the oil for the first time at the recommended interval.

2. Take oil samples at the end of each PM interval before PM and again after PM. Starting and ending samples provide a baseline for how much the oil is degrading over the PM period.

3. If particulate counts are higher than ISO 18/15, off-board filtration (kidney-looping) should be used to clean the oil.

4. The use of Ultra-High Efficiency (UHE) filters is recommended for most applications. These filters trap and retain more debris and help to maintain higher levels of oil cleanliness. However, UHE filters may plug during the PM intervals as they clean the system. This is not abnormal and is not a filter failure. Once the system can maintain ISO 18/15 or better at the end of the PM period, the UHE filters are maintaining cleanliness and off-board filtration may be discontinued if desired. Typically, once oil cleanliness achieves ISO 16/13 or better at the end of the PM period, off-board filtration will not clean the oil any further.

5. After the first oil change, oil life may be extended based on two conditions. Develop a plan to monitor these conditions with the SOS lab.
    a. Particulate levels should be ISO 18/15 or less.
    b. Oxidation should not exceed 80%. Oxidation typically increases at a fairly predictable rate until it reaches around 80%, then increases very rapidly. Oil should be changed before oxidation reaches 80%.

4.0 Benefits

An increasing number of customers have extended oil life by as much as 400% in some cases. This extension results in fewer oil changes and significant savings in new oil and disposal costs of used oil.
Examples include:

785 – 793 Trucks	Recommended Change Interval (Hours)	Actual Change Interval (Hours)
Rear Axle	2,000	8,000
Transmission	1,000	2,000
Steering	2,000	3,000
Hydraulics	2,000	3,000

The actual savings may vary somewhat depending on the amount of extended life achieved, new oil cost, and disposal cost.

5.0 Resources Required

1. Sampling and oil life extension plan with SOS facility.

2. Off-board filtration equipment and training in place.

3. Personnel trained and assigned to monitor oil conditions and make decisions to continue to run or change the oil based on oil condition.

6.0 Supporting Attachments / References

Improving Component Durability booklets (see next page)
Specifically:
Fluid Cleanliness Management SEBF1020

7.0 Related Best Practices

0806-2.1-1000	Fluid cleanliness management
0806-2.1-1003	On-board Machine Filtration

8.0 Acknowledgements

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Resources were used from several large customers, however specific details were not available for sharing.